

TCMC · HARMONY EARS

Harmonic Seventh Pitch Trainer

User Manual

(七和弦泛音音準練習)

App Version: v1.0

Producer Ray Chu (朱元雷)
Taiwan Choral Music Center (TCMC)

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1. Software Overview

“Harmonic Seventh Pitch Trainer” is a web-based ear-training tool developed by the Taiwan Choral Music Center (TCMC) that extends the training approach of the companion “Harmonic Third Pitch Trainer,” helping singers tune the seventh of major-seventh and dominant-seventh chords using Just Intonation (JI) as the reference standard.

The app plays the root, the harmonic third (major third, -13.7 cents in just intonation), and the harmonic fifth (perfect fifth, $+2$ cents) in sequence, gradually building up a four-part harmony, then plays the target harmonic seventh (major seventh, -11.7 cents, or dominant/minor seventh, -31.2 cents). It then activates the microphone to detect the pitch the user sings in real time, displaying graphically whether the pitch falls within a ± 6 -cent tolerance band.

Key Features

- Switch between major-seventh and dominant-seventh chords; root notes cover all 12 keys (arranged by circle of fifths)
- Root $\rightarrow$ third $\rightarrow$ fifth $\rightarrow$ seventh cumulative playback, gradually building the full four-part harmony
- Switch between female voice and male voice (automatic octave drop)
- Real-time microphone pitch detection, with color (green/red) indicating instantly whether the pitch is accurate
- Three independent toggles — play chord, play seventh, microphone — that can be combined freely for different practice modes
- Chinese / English interface toggle, switchable anytime from the top-right corner
- All reference tones are pure sine waves, clean in pitch with no extra harmonics
- Built-in anti-noise-misjudgment mechanism, plus a widened pitch-detection search range tailored to the seventh's higher register, preventing the reading from jumping back near the root when sung slightly sharp

 *Usage tip: we recommend practicing with wired headphones or a phone's built-in microphone. Some Bluetooth headsets automatically switch to a lower-quality call-audio mode when “playing and recording simultaneously,” which can affect the stability of pitch detection.*

2. Interface Overview

Chord Type

Choose whether to practice a “Major 7th Chord” (just intonation $15:8$, -11.7 cents) or a “Dominant 7th Chord” (just intonation $7:4$, -31.2 cents) for this session. The root, third, and fifth are the same in both modes — only the seventh's target pitch differs.

Voice Part

“Female Voice (original key)” sings directly in the selected key; “Male Voice (down an octave)” automatically lowers the root, third, fifth, and seventh by one octave, allowing male singers to practice using the same set of keys.

Key (Root Note)

Root-note buttons for all 12 keys, arranged in the order: C4, F4, Bb3, Eb4, Ab3, Db4, Gb3, B3, E4, A3, D4, G3 (circle-of-fifths order).

Playback Controls

- ▶ Start / ■ Stop: begins or ends the entire practice sequence
- ▶ Play Chord / ■ Stop Chord: fully plays all four notes — root, third, fifth, and seventh
- ▶ Play Seventh / ■ Stop Seventh: plays only the seventh as a standalone reference tone
- ▶ Microphone On / ■ Microphone Off: manually controls whether the user's voice is being detected

Language Toggle

The “EN / 中文” button in the top-right corner switches the interface language at any time; all labels, instructions, and playback status messages update to match, with no page reload required.

Pitch Graph

The real-time pitch graph at the bottom of the screen displays four reference lines (root, third, fifth, and target seventh) along with the user's current pitch trace in real time. See Chapter 4, “Pitch-Accuracy Principle,” for details.

3. Steps to Use

[Click here to open: Harmonic Seventh Pitch Trainer \(GitHub Pages\)](#)

1. Under “Chord Type,” choose Major 7th Chord or Dominant 7th Chord.
2. Under “Voice Part,” choose Female Voice or Male Voice.
3. Under “Key (Root Note),” select the key you want to practice.
4. Press “▶ Start.” The app will automatically play, in order: root → add the third (two-part harmony) → add the fifth (three-part harmony) → play the target seventh.
5. Once the demo playback ends, the microphone turns on automatically and starts listening — begin singing the seventh, using Hm, Oo, Ah, or any syllable you like.
6. Sing the seventh and watch your live pitch line on the graph (bright green = within the ± 6 -cent tolerance band; red = outside it) to see whether it aligns with the gold target line for the seventh.

7. If you'd like to sing along with the accompaniment (headphones recommended), toggle “Play Chord” or “Play Seventh”; if you don't need microphone detection, press “■ Microphone Off” to pause it.
8. When you're done, press “■ Stop,” or simply switch the key, voice part, or chord type to start the next round.

💡 The first time you use the app, your browser will prompt for microphone permission — choose Allow, or real-time pitch detection won't work.

💡 Mobile browsers impose stricter limits on audio playback, requiring one manual tap before playback can be unlocked. If there's no sound the first time you press Start, pressing it again should work.

4. Pitch-Accuracy Principle: Just Intonation vs. Equal Temperament

Fixed-pitch instruments such as the modern piano and guitar are almost universally tuned to “Equal Temperament” (ET), which divides an octave into 12 equal semitones for easy modulation between keys. Except for the octave itself, every other interval in ET differs slightly from the simplest whole-number ratios found in “Just Intonation” (JI). Because a seventh chord adds one more layer beyond the triad, its interval structure is more complex, and the gap between just intonation and ET is especially pronounced on the seventh.

In a cappella or voice-led choral singing, aligning intervals to the whole-number ratios of just intonation produces a more blended sound with almost no beating — which is why this app uses just intonation as its pitch-training target.

Just Intonation Ratios of the Four Notes

- Root: ratio 1:1, the reference pitch, identical to ET
- Major Third (the third): ratio 5:4, about −13.7 cents lower than ET
- Perfect Fifth (the fifth): ratio 3:2, about +2 cents higher than ET
- Major Seventh (the major-7th chord's seventh): ratio 15:8, about −11.7 cents lower than ET
- Minor/Dominant Seventh (the dominant-7th chord's seventh): ratio 7:4, about −31.2 cents lower than ET

A “cent” is the unit musicians use to measure fine differences in pitch; each ET semitone is fixed at exactly 100 cents. The dominant-seventh chord's seventh has a gap of −31.2 cents — nearly a third of a semitone — the largest gap of the four chord tones between just intonation and ET, and the one that requires the most deliberate practice to sing accurately.

A Closer Look: Three Just-Intonation Versions of the Minor/Dominant Seventh

In harmony theory, the “seventh” of a dominant or minor-seventh chord doesn't have a single absolute just-intonation ratio — depending on musical style and harmonic context, at least three versions are commonly used:

- 7:4 (the harmonic seventh): about -31.2 cents from ET. Built from the complete 4:5:6:7 overtone-series ratio for the root, third, fifth, and seventh, this is the simplest relationship among the four tones and produces the least beating. It's the target most often used in a cappella harmony — especially barbershop — when pursuing a fully “locked,” fused chord.
- 16:9 (the Pythagorean minor seventh): about -3.9 cents from ET, very close to equal temperament. Built by stacking four pure perfect fifths, it tends to appear in contexts that emphasize a chain of fifths, strong melodic/linear motion, or where the seventh functions more as a passing tone than a vertically fused harmony.
- 9:5 (the 5-limit minor seventh): about $+17.6$ cents from ET. Derived by stacking a perfect fifth (3:2) with a minor third (6:5) — in other words, treating the seventh as “a minor third above the fifth.” This version is common in some 5-limit choral tuning systems, used when the seventh functions harmonically as an extension of the fifth.

💡 *This app uses the 7:4 version as its practice target, since it most closely matches the barbershop harmony aesthetic and produces the most fused, least-beating sound when all four notes sound together. If the repertoire you're working on leans more toward a chain-of-fifths or 5-limit tuning system, 7:4 isn't necessarily the only “correct” answer — feel free to adjust your judgment to fit the actual musical style.*

💡 *Pitch tolerance: this app judges a pitch as “in tune” within ± 6 cents of the seventh's target — any pitch within that range is treated as aligned with the just-intonation seventh.*

5. Appendix: 12-Key Seventh-Chord Frequency Reference Tables

The two tables below list, for all 12 keys (in circle-of-fifths order), the actual Hz values of the third, fifth, and seventh under Equal Temperament (ET, A4 = 440 Hz) and Just Intonation (JI), provided as a teaching and tuning reference. The root itself is defined against ET's A4 = 440 Hz, while the third, fifth, and seventh are derived from the root using just-intonation whole-number ratios (major third 5:4, perfect fifth 3:2, major seventh 15:8, minor seventh 7:4).

💡 *All Hz values below are for “Female Voice (original key)” pitch; for Male Voice (down an octave) practice, divide every value by 2 to get the actual frequency.*

Major 7th Chord Frequency Table (12 Keys)

Root	Root Hz	3rd ET	3rd JI	5th ET	5th JI	7th ET	7th JI
C4	261.63	329.63	327.03	392.00	392.44	493.88	490.55
F4	349.23	440.00	436.54	523.25	523.84	659.26	654.80
Bb3	233.08	293.66	291.35	349.23	349.62	440.00	437.03
Eb4	311.13	392.00	388.91	466.16	466.69	587.33	583.36
Ab3	207.65	261.63	259.57	311.13	311.48	392.00	389.35
Db4	277.18	349.23	346.48	415.30	415.77	523.25	519.72
Gb3	185.00	233.08	231.25	277.18	277.50	349.23	346.87
B3	246.94	311.13	308.68	369.99	370.41	466.16	463.02
E4	329.63	415.30	412.03	493.88	494.44	622.25	618.05
A3	220.00	277.18	275.00	329.63	330.00	415.30	412.50
D4	293.66	369.99	367.08	440.00	440.50	554.37	550.62
G3	196.00	246.94	245.00	293.66	294.00	369.99	367.50

💡 The seventh's cent difference is calculated from "JI Hz relative to ET Hz." The major third's difference is -13.7 cents, the fifth's is $+2$ cents, and the major seventh's is -11.7 cents.

Dominant 7th Chord Frequency Table (12 Keys)

Root	Root Hz	3rd ET	3rd JI	5th ET	5th JI	7th ET	7th JI
C4	261.63	329.63	327.03	392.00	392.44	466.16	457.84
F4	349.23	440.00	436.54	523.25	523.84	622.25	611.15
Bb3	233.08	293.66	291.35	349.23	349.62	415.30	407.89
Eb4	311.13	392.00	388.91	466.16	466.69	554.37	544.47
Ab3	207.65	261.63	259.57	311.13	311.48	369.99	363.39
Db4	277.18	349.23	346.48	415.30	415.77	493.88	485.07
Gb3	185.00	233.08	231.25	277.18	277.50	329.63	323.75
B3	246.94	311.13	308.68	369.99	370.41	440.00	432.15
E4	329.63	415.30	412.03	493.88	494.44	587.33	576.85
A3	220.00	277.18	275.00	329.63	330.00	392.00	385.00
D4	293.66	369.99	367.08	440.00	440.50	523.25	513.91
G3	196.00	246.94	245.00	293.66	294.00	349.23	343.00

💡 The seventh's cent difference is calculated from "JI Hz relative to ET Hz." The major third's difference is -13.7 cents, the fifth's is $+2$ cents, and the minor/dominant seventh's is -31.2 cents.

6. Troubleshooting

Microphone not responding / can't detect pitch

- Confirm your browser has allowed this page to use the microphone (check the permission icon in the address bar)
- If you're using an in-app or in-chat preview window on a mobile browser, security restrictions may prevent access to the microphone — open the URL directly in a standard mobile or desktop browser (Chrome, Safari, etc.) instead
- Confirm the URL starts with HTTPS; a plain local file (file://) often can't get microphone access on mobile browsers

No sound the first time you press "Start"

- This is caused by mobile browsers' audio playback restrictions — pressing "Start" again usually resolves it
- Confirm your device isn't muted and the browser tab hasn't been silenced by the system

Singing the seventh gets misjudged as jumping back to the root

- The app has already widened its pitch-detection search range to account for the seventh's higher register — even if you sing slightly sharp, it should usually still be detected correctly instead of snapping back near the root

- If this keeps happening, try slowing down to first confirm the seventh's target pitch before singing along, and make sure the environment is quiet with a stable recording distance

The pitch line cuts out or jumps around

- The app has a built-in anti-noise-misjudgment mechanism: brief detection gaps (around 0.5 seconds) are automatically bridged, so brief noise or breath sounds aren't mistaken for pitch jumps
- Avoid using a Bluetooth headset's microphone — simultaneous Bluetooth playback and recording noticeably degrades audio quality; wired headphones or the phone's built-in microphone are recommended instead

Can't hear "Play Chord" or "Play Seventh" clearly

- “Play Chord” plays the root, third, fifth, and seventh together; “Play Seventh” plays only the seventh by itself — the two are independent and can be turned on separately or together
- Once the listening phase begins, the auto-played accompaniment is muted by default — you'll need to press the playback toggle yourself to hear it

7. Version Information

Current version: v1.0

Production Team

Produced by: Taiwan Choral Music Center (TCMC)

Producer: Ray Chu (朱元雷)

Nature of software: a web-based tool for choral instruction and self-practice, and an extension of the “Harmonic Third Pitch Trainer.”